

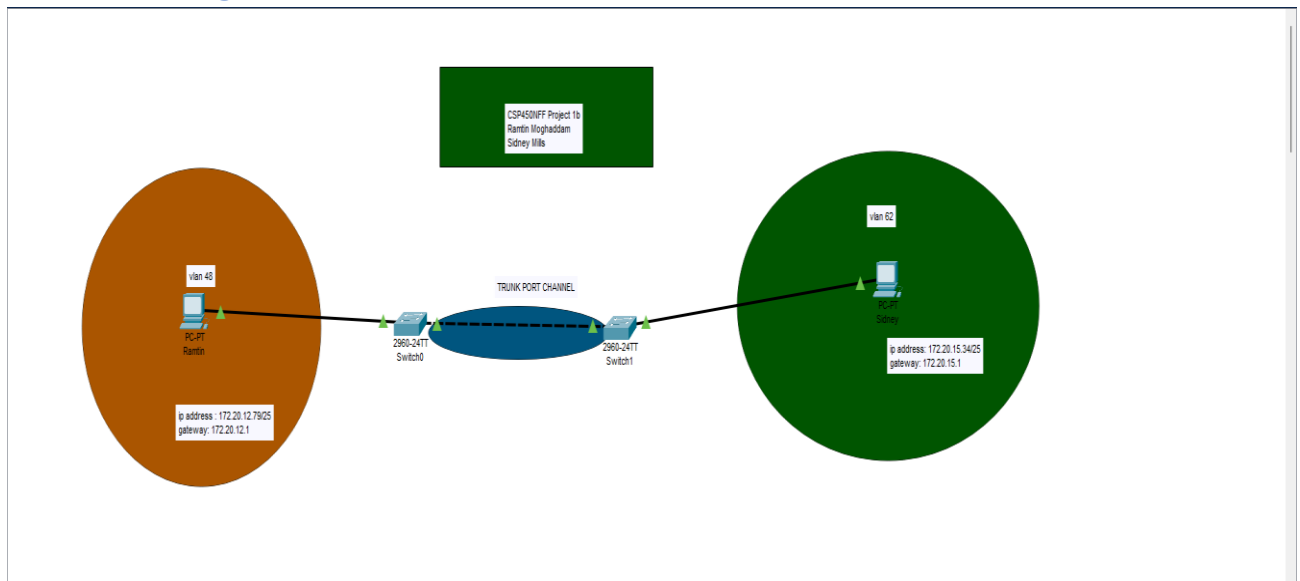
Ramtin Moghaddam

Project Finished together with **Sidney Mills**

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1. Network Diagram:



2. The script Used to configure the switch

2960 switch (Ramtin):

```
config t
```

```
Vlan 62
```

```
    Name vlan 62
```

```
    Exit
```

```
Vlan 48
```

```
    Name vlan 48
```

```
    Exit
```

```
Interface 1/1/5
```

```
    no shutdown
```

```
    no routing
```

```
    vlan trunk allowed 48,62
```

```
    exit
```

```
interface 1/1/6
```

```
    no shutdown
```

```
    no routing
```

```
    vlan trunk allowed 48,62
```

```
    exit
```

```
interface 1/1/1
```

```
    no shutdown
```

```
    no routing
```

```
    vlan access 48
```

```
    exit
```

```
int vlan 48
```

```
    ip address 172.20.12.1 255.255.255.128
```

```
    exit
```

```
int vlan 62
```

```
    ip address 172.20.15.1 255.255.255.128
```

```
int lag 1
```

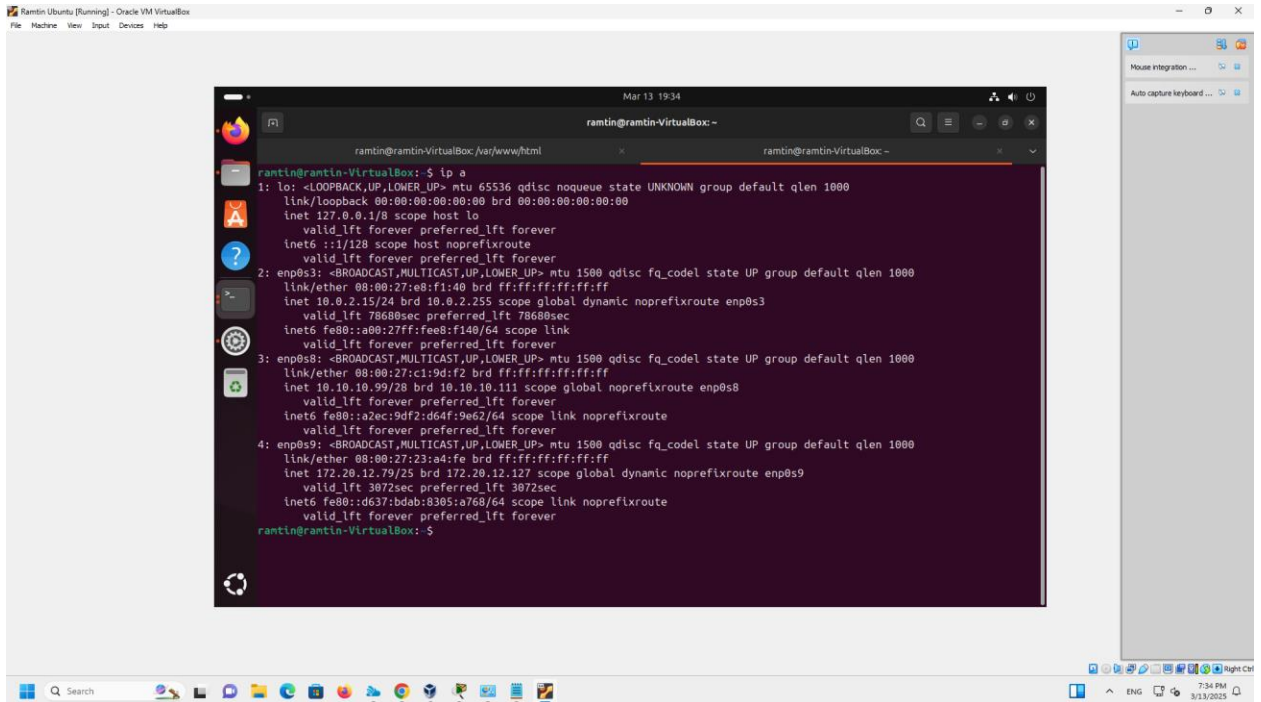
```
    no shut
```

```
no route
vlan trunk allowed all
lACP mode active
dhcp-server vrf default
pool Ramtin
    range 172.20.12.2 172.20.12.126
    default-router 172.20.12.1
    exit
pool Sidney
    range 172.20.15.2 172.20.15.126
    default-router 172.20.15.1
    exit
authoritative
enable
```

2530 switch (Sidney):

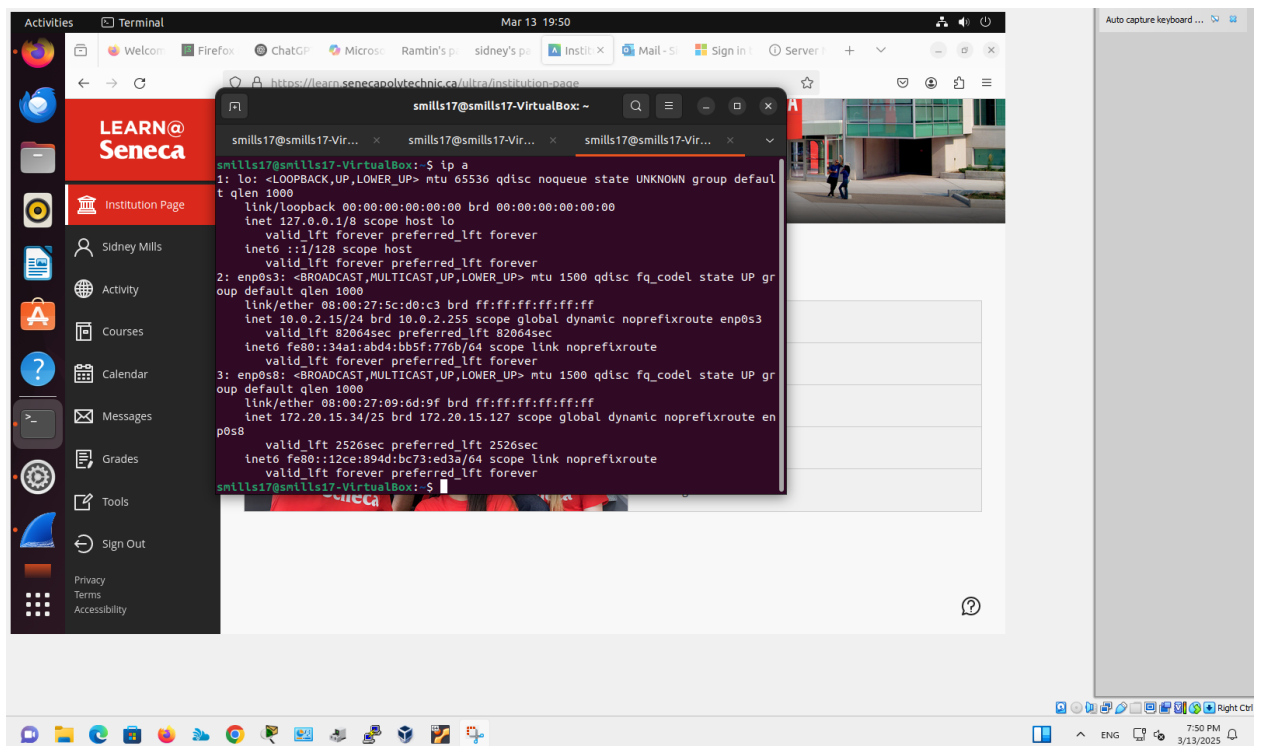
```
Vlan 1
    name "DEFAULT_VLAN"
    no untagged 1
    untagged 4-28, Trk1
    ip address 10.10.10.88 255.255.255.240
    exit
vlan 62
    name "VLAN62"
    untagged 1
    tagged Trk1
    no ip address
    exit
spanning-tree Trk1 priority 4
```

3. Ip a



The screenshot shows a terminal window titled "ramtin@ramtin-VirtualBox: ~" with the command "ip a" executed. The output displays network interface details for four interfaces: lo, enp0s3, enp0s8, and enp0s9. Each interface shows its link/ether address, IP address, netmask, and other configuration details.

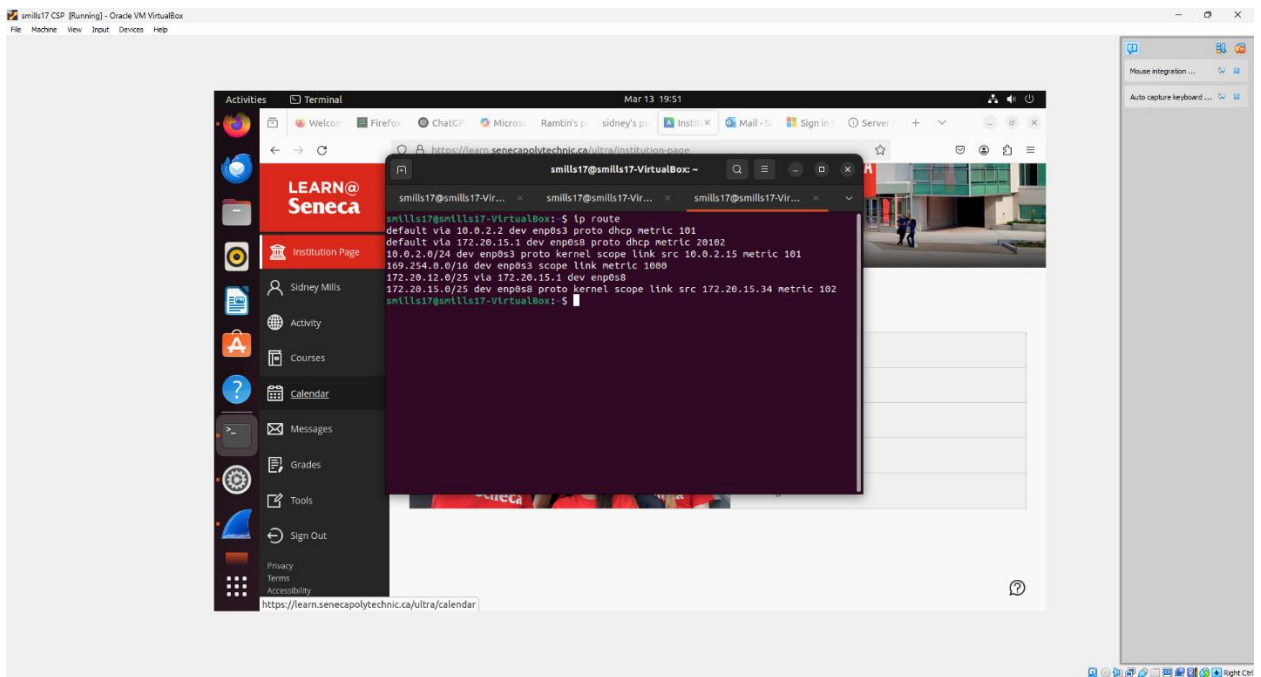
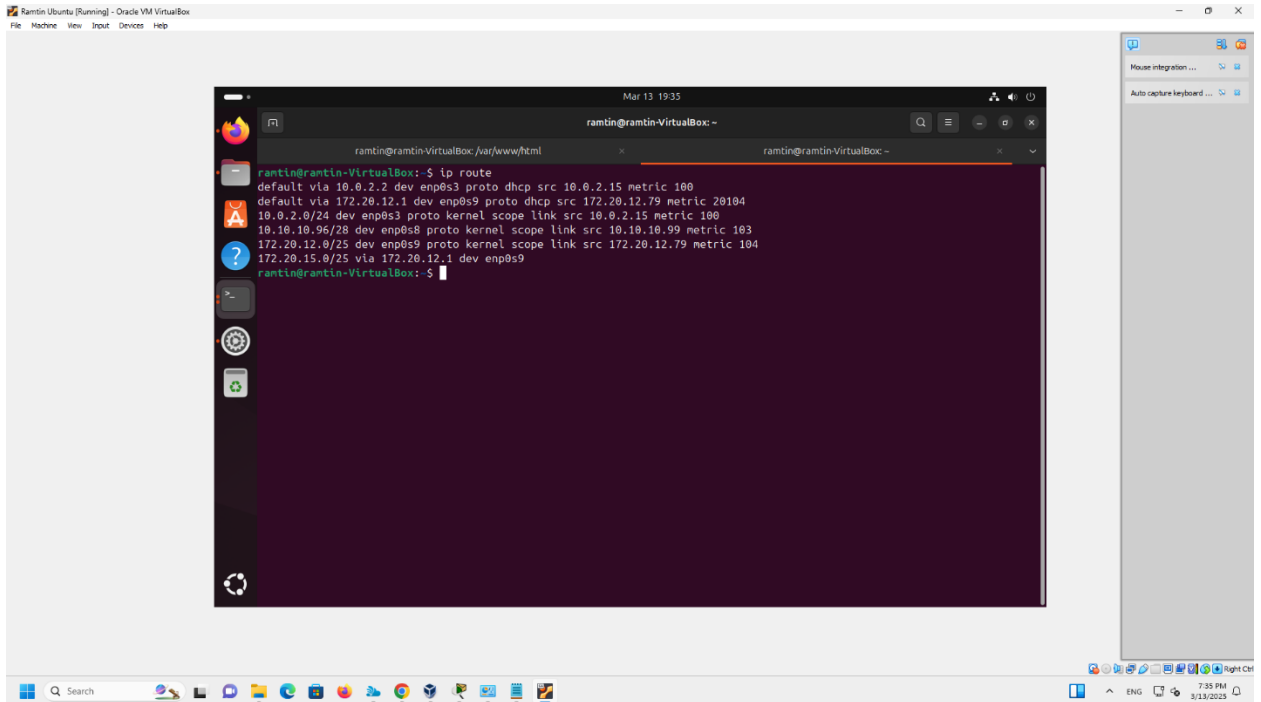
```
ramtin@ramtin-VirtualBox: ~  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:e8:f1:40 brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3  
        valid_lft 78680sec preferred_lft 78680sec  
    inet6 fe80::a00:27ff:fe08:f140/64 scope link  
        valid_lft forever preferred_lft forever  
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:c1:9d:f2 brd ff:ff:ff:ff:ff:ff  
    inet 10.10.10.99/28 brd 10.10.10.111 scope global noprefixroute enp0s8  
        valid_lft forever preferred_lft forever  
    inet6 fe80::a2ec:9df2:d64f:9e62/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
4: enp0s9: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:23:a4:fe brd ff:ff:ff:ff:ff:ff  
    inet 172.20.12.79/25 brd 172.20.12.127 scope global dynamic noprefixroute enp0s9  
        valid_lft 3072sec preferred_lft 3072sec  
    inet6 fe80::d637:bdab:8305:a768/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
ramtin@ramtin-VirtualBox: ~
```



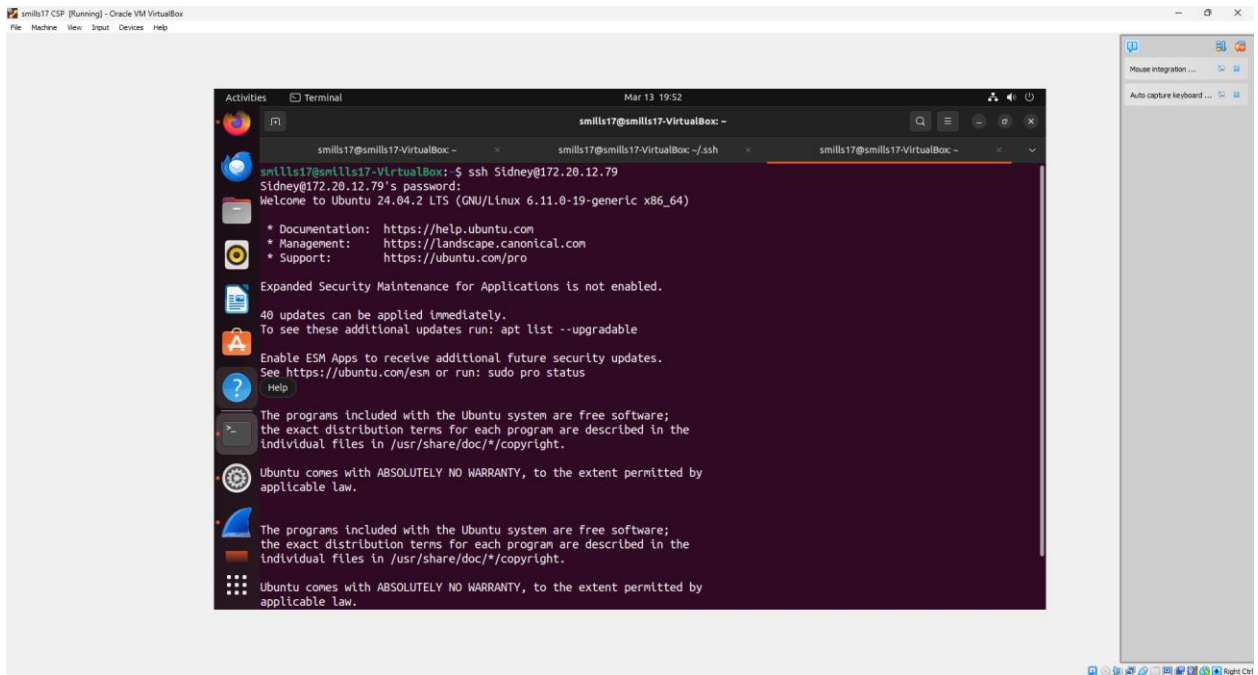
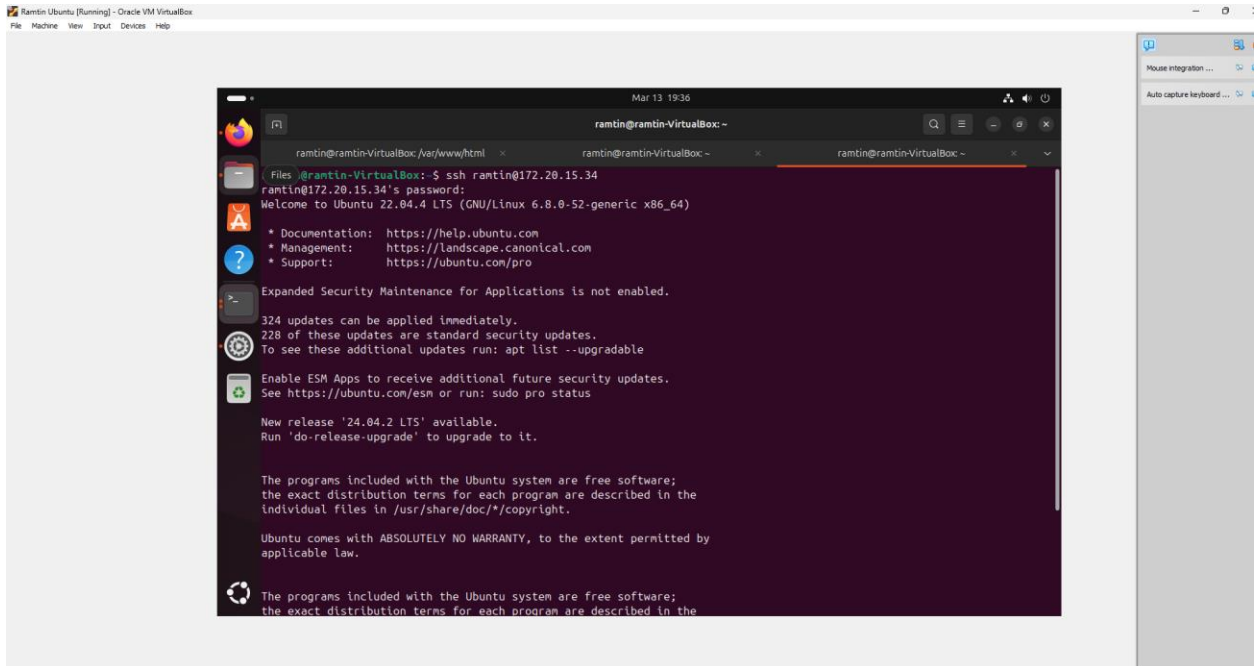
The screenshot shows a terminal window titled "smills17@smills17-VirtualBox: ~" with the command "ip a" executed. The output displays network interface details for four interfaces: lo, enp0s3, enp0s8, and enp0s9. Each interface shows its link/ether address, IP address, netmask, and other configuration details. The terminal window is overlaid on a web browser window showing the "LEARN@ Seneca" website.

```
smills17@smills17-VirtualBox: ~  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:5c:d0:c3 brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3  
        valid_lft 82064sec preferred_lft 82064sec  
    inet6 fe80::34a1:abd4:b05f:776b/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:09:6d:9f brd ff:ff:ff:ff:ff:ff  
    inet 172.20.15.34/25 brd 172.20.15.127 scope global dynamic noprefixroute enp0s8  
        valid_lft 2526sec preferred_lft 2526sec  
    inet6 fe80::12ce:894d:bc73:ed3a/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
smills17@smills17-VirtualBox: ~
```

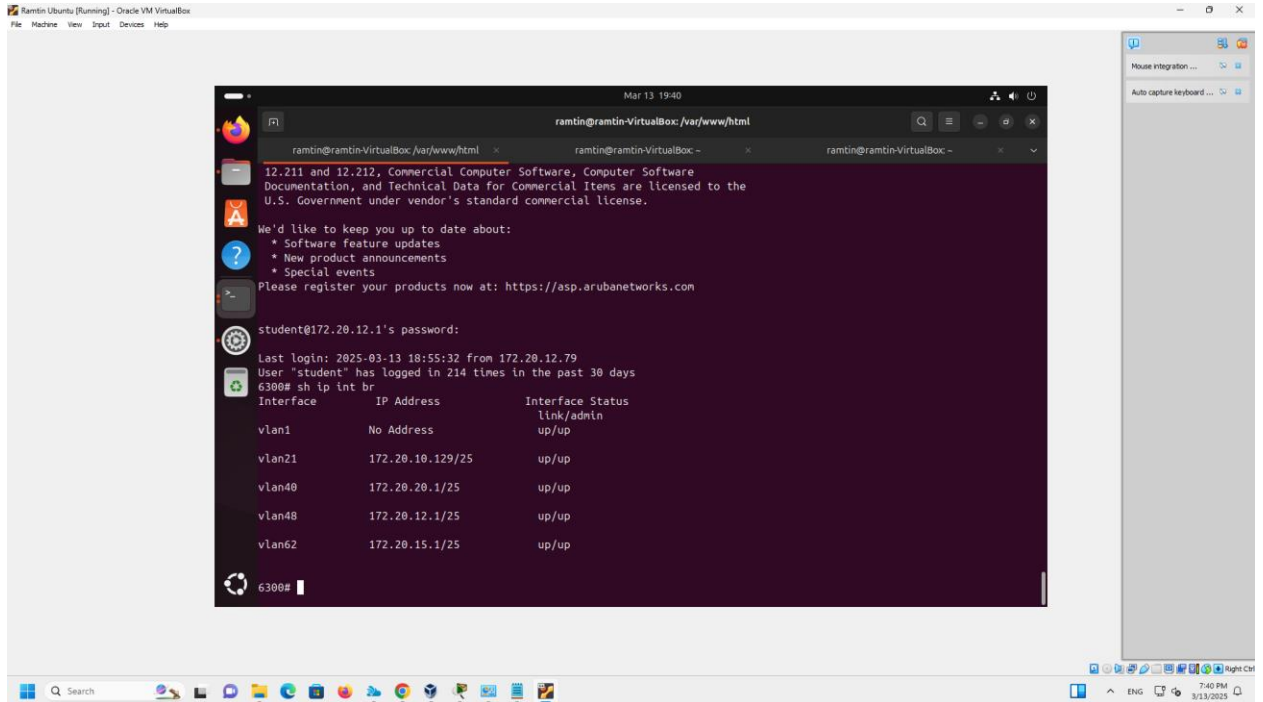
4. Ip route



5. Successfully SSH'ed to other VM



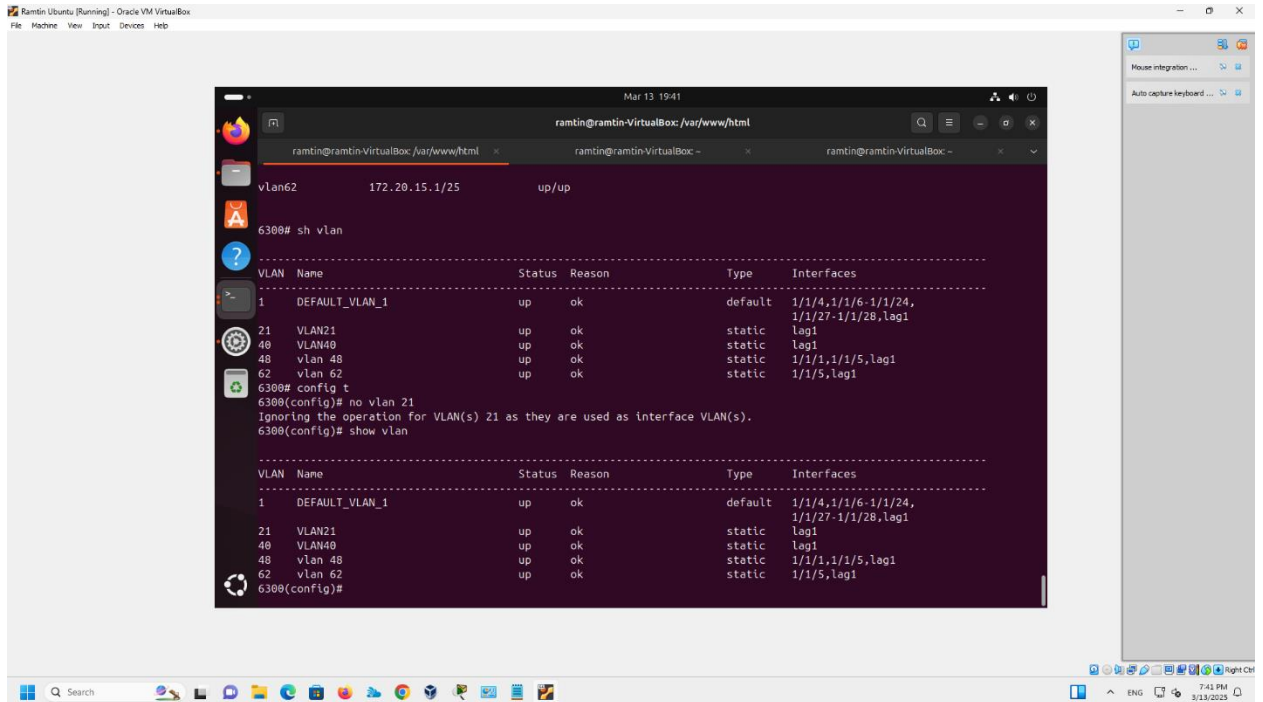
6. sh ip int br



The screenshot shows a terminal window within a VM. The user has entered the command `sh ip int br`. The output displays a table of interfaces and their IP addresses.

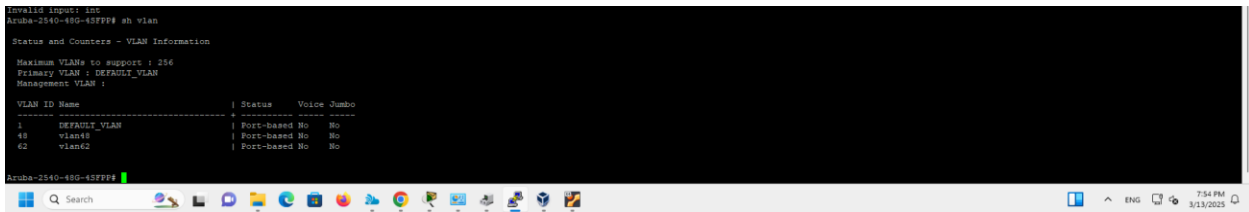
Interface	IP Address	Interface Status
vlan1	No Address	link/admin up/up
vlan21	172.20.10.129/25	up/up
vlan40	172.20.20.1/25	up/up
vlan48	172.20.12.1/25	up/up
vlan62	172.20.15.1/25	up/up

7. show vlan

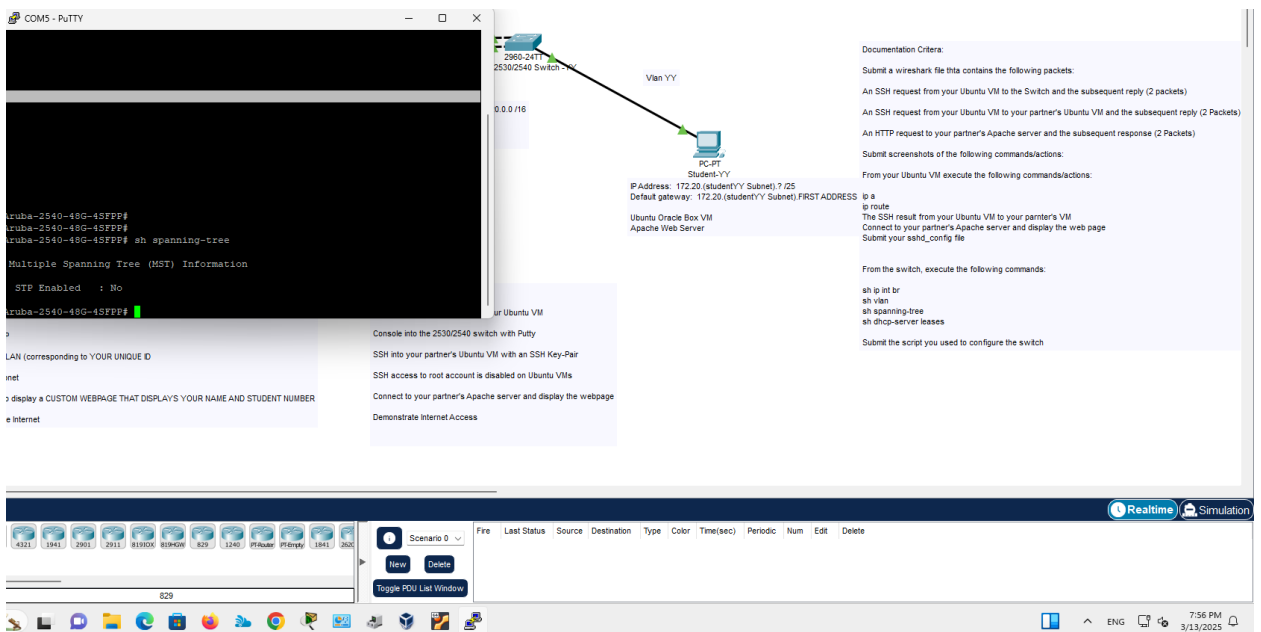
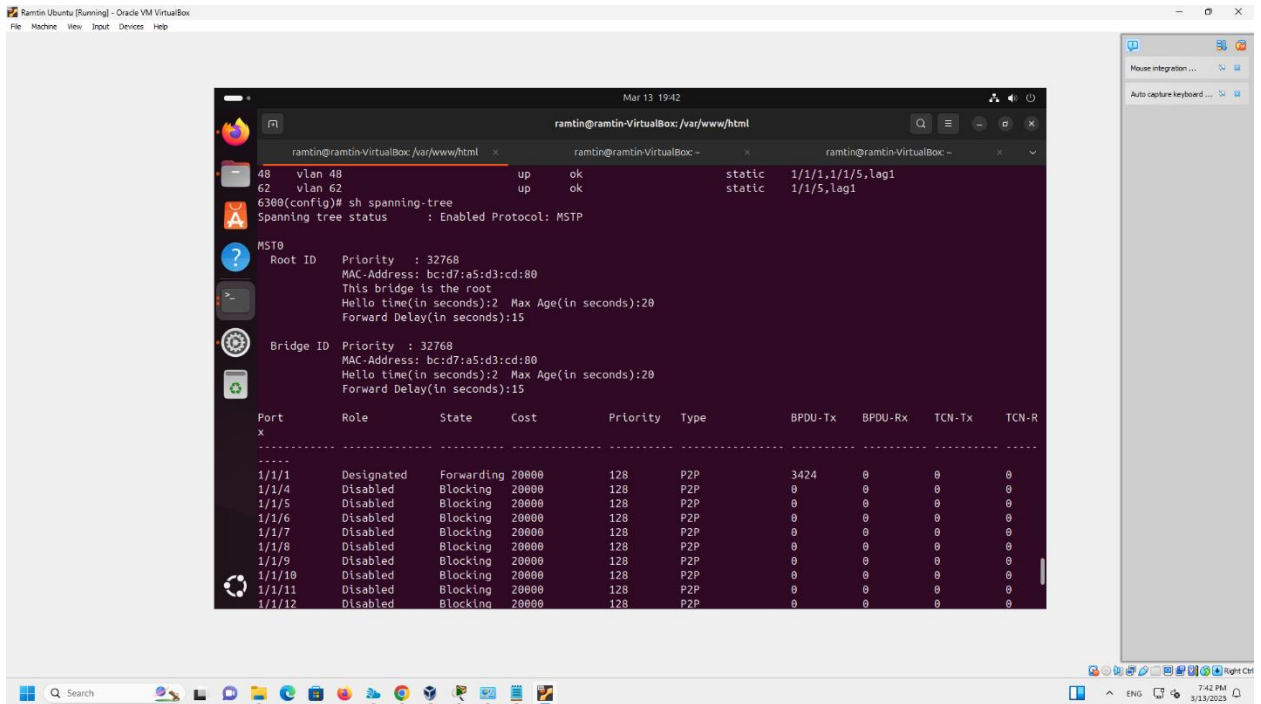


The screenshot shows a terminal window within a VM. The user has entered the command `show vlan`. The output displays a table of VLANs and their configurations.

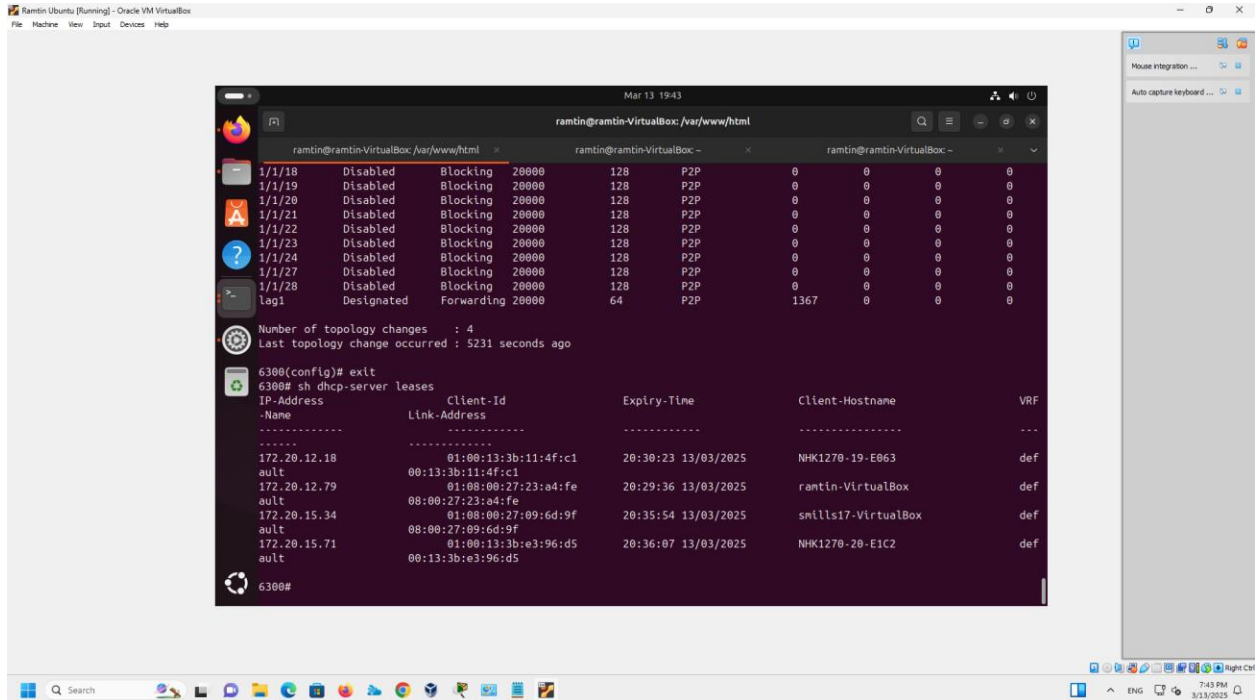
VLAN	Name	Status	Reason	Type	Interfaces
1	DEFAULT_VLAN_1	up	ok	default	1/1/4,1/1/6-1/1/24, 1/1/27-1/1/28,lag1
21	VLAN21	up	ok	static	lag1
40	VLAN40	up	ok	static	lag1
48	vlan 48	up	ok	static	1/1/1,1/1/5,lag1
62	vlan 62	up	ok	static	1/1/5,lag1



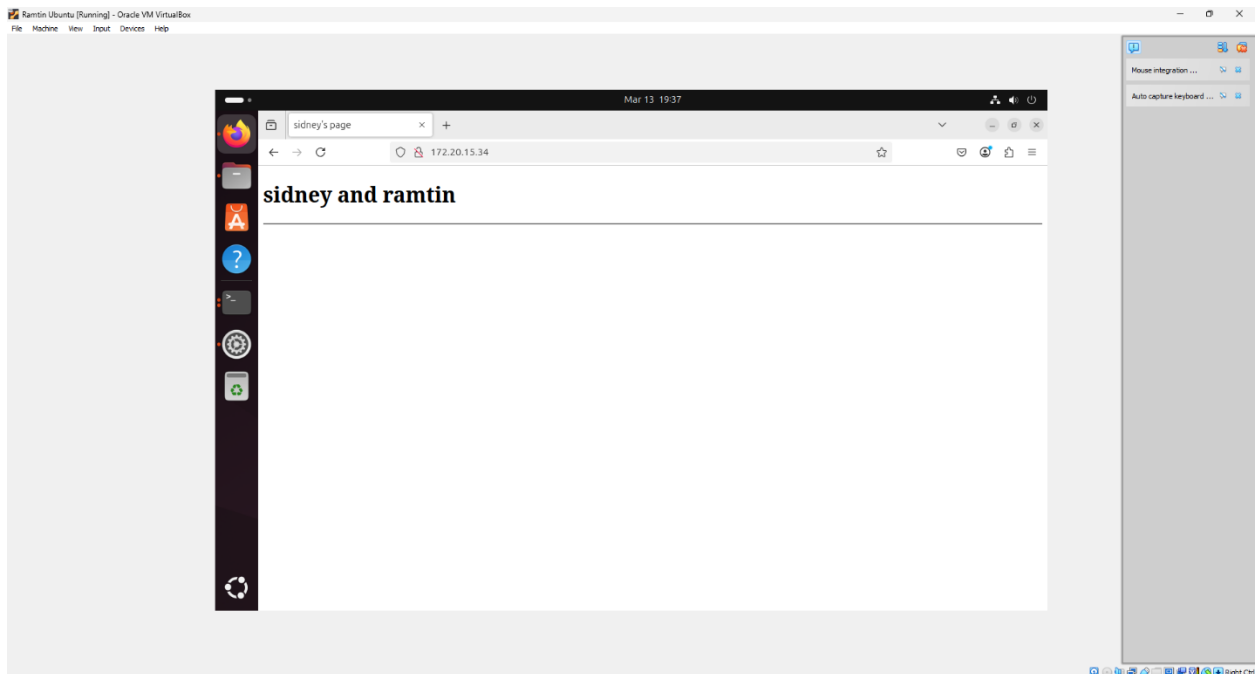
8. show spanning-tree

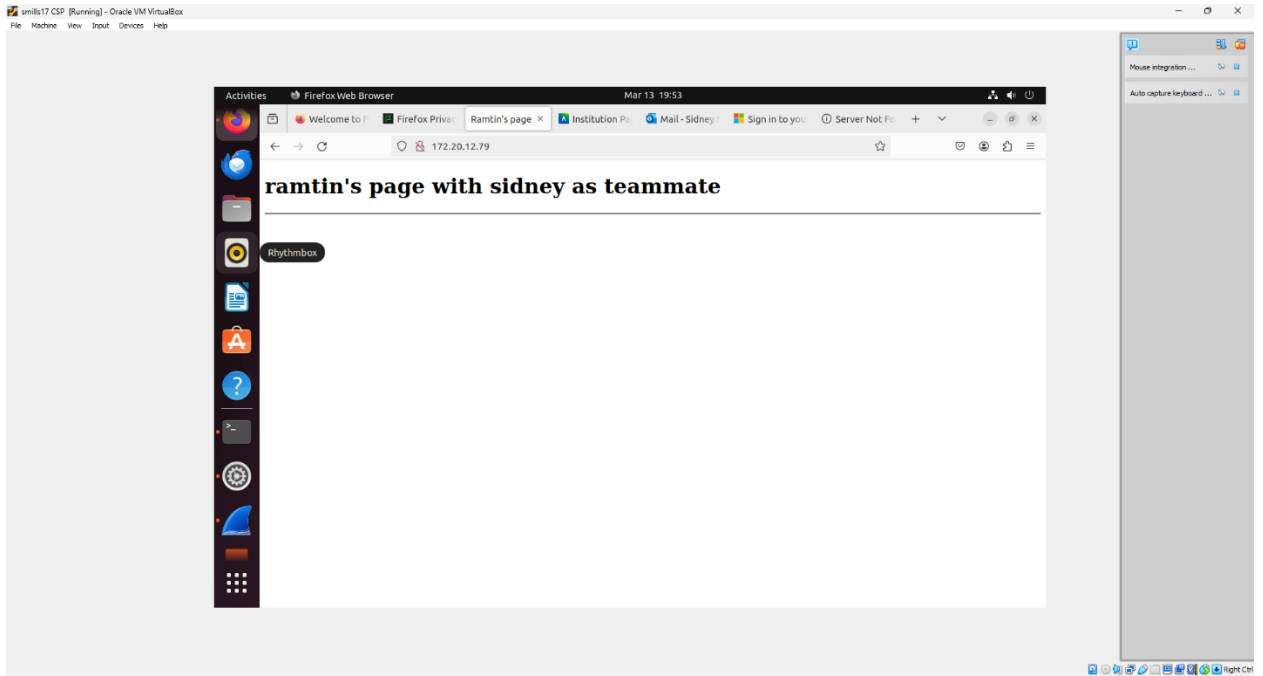


9. sh dhcp-server leases



10. webpage





11. wireshark capture

Wireshark capture showing network traffic between 172.20.12.79 and 172.20.12.1. The capture is filtered by 'Apply a display filter: <Ctrl>F'.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	172.20.12.79	172.20.12.1	TCP	76	60368 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=2043652596 TSecr=0 WS=128
2	0.000630641	172.20.12.1	172.20.12.79	TCP	76	22 → 60368 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=1460 SACK_PERM=1 TSval=3525520032 TSecr=2043652596 WS=128
3	7.626672760	172.20.12.79	172.20.15.34	TCP	76	45550 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=1199386177 TSecr=0 WS=128
4	7.627407574	172.20.15.34	172.20.12.79	TCP	76	22 → 45550 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM=1 TSval=1441640013 TSecr=1199386177 WS=128
5	16.550243424	172.20.12.79	172.20.15.34	TCP	76	55526 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=1199395100 TSecr=0 WS=128
6	16.550904916	172.20.15.34	172.20.12.79	TCP	76	80 → 55526 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM=1 TSval=1441648937 TSecr=1199395100 WS=128

Frame 1: 76 bytes on wire (608 bits), 76 bytes captured (608 bits) on interface any, id 0
 Linux cooked capture v1
 Internet Protocol Version 4, Src: 172.20.12.79, Dst: 172.20.12.1
 Transmission Control Protocol, Src Port: 60368, Dst Port: 22, Seq: 0, Len: 0

```

0000  00 04 00 01 00 06 08 00 27 23 a4 f6 00 00 08 00  ....'8.....
0010  45 10 00 3c 32 f8 40 00 40 06 97 3b ac 14 0c 4f  E<<2@ @.;...0
0020  ac 14 0c 01 eb d0 00 16 d9 6a e3 b8 00 00 00 00  ....-j.....
0030  a0 02 fa f0 70 a7 00 00 02 04 05 b4 04 02 08 0a  ....p.....
0040  79 cf a9 f4 00 00 00 01 03 03 07  y.....
  
```

lab1_2.pcapng						
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help						
[Apply a display filter: <Ctrl>]						
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	172.20.15.34	172.20.12.79	TCP	74	46216 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=1440422518 TSecr=0 WS=128
2	0.000072757	172.20.12.79	172.20.15.34	TCP	74	80 → 46216 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM=1 TSval=1198168680 TSecr=1440422518 WS=128
3	326.431899903	172.20.12.79	172.20.15.34	TCP	74	43510 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=1198495110 TSecr=0 WS=128
4	326.431984665	172.20.15.34	172.20.12.79	TCP	74	22 → 43510 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM=1 TSval=1440748950 TSecr=1198495110 WS=128
5	1056.5551167...	172.20.15.34	172.20.12.79	TCP	74	37416 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=1441479074 TSecr=0 WS=128
6	1056.5557865...	172.20.12.79	172.20.15.34	TCP	74	22 → 37416 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM=1 TSval=1199225237 TSecr=1441479074 WS=128